

- 3 (a) (i) Diffraction is the bending and spreading of waves through the aperture or around an obstacle.
- (ii) Phase difference between 2 waves meeting at a point is an angular measure of the difference in their states of motion as a fraction of a complete cycle, expressed in degrees or radians, where one complete cycle is equivalent to 360° .
- (iii) Two wave sources possess coherence if the waves emerging from them have a constant phase difference.

Note: Do not suggest coherent as same phase!

- (b) (i) Assuming emitters are coherent and emit waves in phase,
At locations AB where their path difference is $n\lambda$, constructive interference resulting in maxima is formed.

At locations where path difference is producing a minima.

Note: A significant number did not give details for conditions $(n + \frac{1}{2})\lambda$, they arrive π radians out of phase resulting in destructive interference for constructive and destructive!
Some gave phase difference as $n\lambda$!

- (ii) 1. The maximum intensity of the fringes increased due to increase in amplitude of the two constructive waves. The minimum intensity remains unchanged. The contrast increases due to a clearer distinction between dark and bright fringes. The position of the fringes are not changed.
- (ii) 2. Changing the phase difference by π radians does not affect the intensity of the fringes but causes the maximum and zero intensity fringes to exchange places.

Note: 1. Candidates states dark fringe would be darker which does not make sense!
2. Please try to understand phase difference and do not suggest pattern disappear.